56.1: Human activities threaten earth's biodiversity

Throughout the biosphere, human activities are altering natural disturbances, trophic structures, energy flow, and chemical cycling—ecosystem processes on which we and all other species depend. We have physically altered nearly half of Earth's land surface, and we use over half of all accessible surface fresh water. In the oceans, stocks of most major fisheries are shrinking because of overharvesting. By some estimates, we may be pushing more species toward extinction than did the asteroid that triggered the Cretaceous mass extinction 66 million years ago (see Figure 25.18).

In this chapter, we'll examine changes happening across Earth, focusing on **conservation biology**, a discipline that integrates ecology, physiology, molecular biology, genetics, and evolutionary biology to conserve the diversity of life on Earth. Efforts to sustain ecosystem processes and stem the extinction of species also connect the life sciences with the social sciences, economics, and humanities.

Extinction is a natural phenomenon that has been occurring since life first evolved; it is the high *rate* of extinction that is of concern. More than 1,000 species have become extinct in the last 400 years, a rate that is 100 to 1,000 times the "background," or typical, extinction rate seen in the fossil record (see Concept 25.4). This comparison indicates that the extinction rate today is high and that human activities threaten Earth's biological diversity at all levels.

Three Levels of Biodiversity

Biodiversity—short for biological diversity—can be considered at three main levels: genetic diversity, species diversity, and ecosystem diversity **(Figure 56.2)**.

Figure 56.2

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Three levels of biodiversity.
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Genetic diversity in a vole population; species diversity in a coastal redwood ecosystem; community and ecosystem diversity across the landscape of an entire region.

The oversized chromosomes in the top diagram symbolize the genetic variation within the population.

Genetic Diversity

Genetic diversity comprises not only the individual genetic variation *within* a population, but also genetic differences *between* populations—differences that often are associated with adaptations to local conditions. If one population becomes extinct, then a species may have lost some of the genetic diversity that makes microevolution possible. This erosion of genetic diversity in turn reduces the adaptive potential of the species.

Species Diversity

Public awareness of the biodiversity crisis centers on species diversity—the number of species in an ecosystem or across the biosphere. To date, scientists have described and formally named about 1.8 million species of organisms. In addition to these named species, many others remain to be discovered: Estimates for the number of species that currently exist range from 5 million to 100 million.

Of particular concern are species that are endangered or threatened. An **endangered species** is one that is in danger of extinction throughout all or much of its range **(Figure 56.3)**, while a **threatened species** is considered likely to become endangered in the near future. The following are just a few statistics that illustrate the problem of species loss:

- As of 2021, according to the International Union for Conservation of Nature and Natural Resources (IUCN), 14% of the 10,000 known species of birds and 26% of the 6,000 known species of mammals are threatened.
- A survey by the Center for Plant Conservation showed that of the nearly 20,000 known plant species in the United States, 200 have become extinct since such records have been kept, and 730 are endangered or threatened.
- In North America, at least 123 freshwater animal species have become extinct since 1900, and hundreds more species are threatened. The extinction rate for North American freshwater fauna is about five times as high as that for terrestrial animals.

Figure 56.3

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A hundred heartbeats from extinction.
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Philippine eagle and Yangtze river dolphin.

These are two members of what Harvard biologist E. O. Wilson calls the Hundred Heartbeat Club, species with fewer than 100 individuals remaining on Earth. The Yangtze River dolphin may be extinct, but a few individuals were reportedly sighted in 2007.

To document that a species has actually become extinct, what factors would you need to consider?

*For suggested answer, see*Appendix A.**

The local populations of a species can also be driven to extinction; for example, populations of a species may be lost in one river system but survive in an adjacent one. Global extinction of a species means that it is lost from *all* the ecosystems in which it lived, leaving them permanently impoverished.

HHMI BioInteractive Video

Surveying Gorongosa's Biodiversity

image

Play

image

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Ecosystem Diversity

The variety of ecosystems on Earth is a third level of biological diversity. Because of the many interactions between members of different species in an ecosystem, the extinction of populations of one species can have a negative impact on other species in the ecosystem (see Figure 54.20). For instance, bats called "flying foxes" are important pollinators and seed dispersers in the Pacific Islands, where they are increasingly hunted as a luxury food **(Figure 56.4)**. Conservation biologists fear that the extinction of flying foxes would also harm the native plants of the Samoan Islands, where four-fifths of the tree species depend on flying foxes for pollination or seed dispersal.

Figure 56.4

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    The endangered Marianas "flying fox" bat (*Pteropus
mariannus*), an important pollinator.
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Marianas flying fox bat eating the flowers on a tree.

Some ecosystems have already been heavily affected by humans, and others are being altered at a rapid pace. Since European colonization, more than half of the wetlands in the contiguous United States have been drained and converted to agricultural and other uses. In California, Arizona, and New Mexico, roughly 90%

of native riparian (streamside) communities have been affected by overgrazing, flood control, water diversions, lowering of water tables, and invasion by non-native plants.

Biodiversity and Human Welfare

Why should we care about the loss of biodiversity? One basic reason concerns our human sense of connection to nature and all forms of life, termed *biophilia*. Moreover, the belief that other species are entitled to life is a pervasive theme of many religions and the basis of a moral argument that we should protect biodiversity. There is also a concern for future human generations. Paraphrasing an old proverb, G. H. Brundtland, a former prime minister of Norway, said, “We must consider our planet to be on loan from our children, rather than being a gift from our ancestors.” In addition to such philosophical and moral justifications, species and genetic diversity bring us many practical benefits.

Benefits of Species and Genetic Diversity

Many threatened species could potentially provide medicines, food, and fibers for human use, making biodiversity a crucial natural resource. Products from aspirin to antibiotics were derived originally from natural sources. If we lose wild populations of plants closely related to agricultural species, we lose genetic resources that could be used to improve crop qualities, such as disease resistance. For instance, in the 1970s, plant breeders responded to devastating outbreaks of the grassy stunt virus in rice (*Oryza sativa*) by screening 7,000 populations of this species and its close relatives for resistance to the virus. One population of a single relative, Indian rice (*Oryza nivara*), was found to be resistant to the virus, and scientists succeeded in breeding the resistance trait into commercial rice varieties. Today, the original disease-resistant population has apparently become extinct in the wild.

In the United States, about 25% of the prescriptions dispensed from pharmacies contain substances originally derived from plants. For example, researchers discovered that the rosy periwinkle, which grows in Madagascar, contains alkaloids that inhibit cancer cell growth **(Figure 56.5)**. This discovery led to treatments for two deadly forms of cancer, Hodgkin’s lymphoma and childhood leukemia, resulting in remission in most cases. Madagascar is also home to five other species of periwinkles, one of which is approaching extinction. Losing these species would mean the loss of any possible medicinal benefits they might offer.

Figure 56.5

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The rosy periwinkle (*Catharanthus roseus*), a plant that
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saves lives.
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Rosy periwinkle in bloom with red flowers.

Each species lost means the loss of unique genes, some of which may code for enormously useful proteins. The enzyme Taq polymerase was first extracted from a bacterium, *Thermus aquaticus*, found in hot springs at Yellowstone National Park. This enzyme is essential for the polymerase chain reaction (PCR) because it is stable at the high temperatures required for automated PCR (see Figure 20.8). DNA from many other species of prokaryotes, living in a variety of environments, is used in the mass production of proteins for new medicines, foods, petroleum substitutes, industrial chemicals, and other products. However, because many species of prokaryotes and other organisms may become extinct before we discover them, we stand to lose the valuable genetic potential held in their unique libraries of genes.

Ecosystem Services

The benefits that individual species provide to humans are substantial, but saving individual species is only part of the reason for preserving ecosystems. Humans evolved in Earth's ecosystems, and we rely on these systems and their inhabitants for our survival. **Ecosystem services** encompass all the processes through which natural ecosystems help sustain human life. Ecosystems purify our air and water. They detoxify and decompose our wastes and reduce the impacts of extreme weather and flooding. The organisms in ecosystems pollinate our crops, control pests, and create and preserve our soils. Moreover, these diverse services are provided for free.

If we had to pay for them, how much would the services of natural ecosystems be worth? In 1997, scientists estimated the value of Earth's ecosystem services at $33 trillion per year, nearly twice the gross national product of all the countries on Earth at the time $18 trillion. It may be more realistic to do the accounting on a smaller scale. For example, New York City invested $1.5 billion to buy land and restore habitat in the Catskill Mountains, the source of much of the city's fresh water. This investment was spurred by increasing pollution of the water by sewage, pesticides, and fertilizers. By harnessing ecosystem services to purify its water naturally, the city avoided spending $2 billion to build new water treatment plants and $300 million a year to run those plants.

There is growing evidence that the functioning of ecosystems, and hence their capacity to perform services, is linked to biodiversity. As human activities reduce biodiversity, we are reducing the capacity of the planet's ecosystems to perform processes critical to our own survival.

Interview

Interview with E. O. Wilson: Advocate for protecting biodiversity

A photograph of E O Wilson, an American biologist.

Threats to Biodiversity

Many different human activities threaten biodiversity on local, regional, and global scales. The threats posed by these activities include four major types: habitat loss, introduced species, overharvesting, and global change.

Habitat Loss

Human alteration of habitat is the single greatest threat to biodiversity throughout the biosphere. Habitat loss has been brought about by factors such as agriculture, urban development, forestry, mining, and pollution. As discussed in more detail later in this chapter, global climate change is also altering habitats today and will have an even larger effect later this century. When no alternative habitat is available or a species is unable to move, habitat loss may mean extinction. The IUCN implicates destruction of habitat as a contributing cause for 73% of the species that have become extinct, endangered, vulnerable, or rare in the last few hundred years.

Habitat loss and fragmentation may occur over immense regions. Approximately 98% of the tropical dry forests of Central America and Mexico have been cut down. The clearing of tropical rain forest in the state of Veracruz, Mexico, mostly for cattle ranching, has resulted in the loss of more than 90% of the original forest, leaving relatively small, isolated patches of forest. Many other natural habitats have also been fragmented by human activities **(Figure 56.6)**.

Figure 56.6

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Habitat fragmentation in the foothills of Los Angeles.
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Houses pack the valleys in the foothills of Los Angeles but the foothills themselves are left undeveloped.

Development in the valleys may confine the organisms that inhabit the narrow strips of hillside.

Habitat fragmentation often reduces the number of species in a region because the smaller populations in habitat fragments have a higher probability of extinction. Prairie covered about 800,000 hectares of southern Wisconsin when Europeans first arrived in North America but occupies only 800 hectares today; most of the original prairie in this area is now used to grow crops. Plant diversity surveys of 54 Wisconsin

prairie remnants conducted in 1948–1954 and 1987–1988 showed that the remnants lost 8–60% of their plant species in the time between the two surveys.

Habitat loss is also a major threat to aquatic biodiversity. About 70% of coral reefs, among Earth's most species-rich aquatic communities, have been damaged by human activities. At the current rate of destruction, 40–50% of the reefs, home to one-third of marine fish species, could disappear in the next 30 to 40 years. Freshwater habitats are also being lost, often as a result of the dams, reservoirs, channel modification, and flow regulation now affecting most of the world's rivers. For example, the more than 30 dams and locks built along the Mobile River basin in the southeastern United States changed river depth and flow. While providing the benefits of hydroelectric power and increased ship traffic, these dams and locks also helped drive more than 40 species of mussels and snails to extinction.

Introduced Species

Introduced species—sometimes called invasive, non-native, or exotic species—are those that humans move intentionally or accidentally from the species' native locations to new geographic regions. Human travel by ship and airplane has accelerated the transplant of species. Free from the predators, herbivores, pathogens, or competitors that limit their populations in their native habitats, such transplanted species may spread rapidly through a new region.

Some introduced species disrupt their new community, often by preying on native organisms or outcompeting native organisms for resources. For example, the brown tree snake was accidentally introduced to the island of Guam from other parts of the South Pacific after World War II, as a "stowaway" in military cargo. Since then, 12 species of birds and 6 species of lizards that the snakes ate have become extinct on Guam. In 1988, the devastating zebra mussel, a suspension-feeding mollusc, was discovered in the Great Lakes of North America, most likely introduced in the ballast water of ships arriving from Europe. Zebra mussels form dense colonies and have disrupted freshwater ecosystems, threatening native aquatic species. They have also clogged water intake structures, causing billions of dollars in damage to domestic and industrial water supplies.

Humans have deliberately introduced many species with good intentions but disastrous effects. An Asian plant called kudzu, which the U.S. Department of Agriculture once introduced in the southern United States to help control erosion, has taken over large areas of the landscape there **(Figure 56.7)**. The European starling was brought intentionally into New York's Central Park in 1890 by a citizens' group intent on introducing all the plants and animals mentioned in Shakespeare's plays. It quickly spread across North America, where its population now exceeds 100 million, displacing many native songbirds.

Figure 56.7

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    Kudzu, an introduced species, thriving in the southeastern
United States.
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Kudzu grows over the walls and roof of a building so that only part of the roof is still visible.

Introduced species are a worldwide problem, contributing to approximately 40% of the extinctions recorded since 1750 and costing billions of dollars each year in damage and control efforts. There are more than 50,000 introduced species in the United States alone.

Overharvesting

The term *overharvesting* refers generally to the harvesting of wild organisms at rates exceeding the ability of their populations to rebound. Species with restricted habitats, such as those that inhabit small islands, are particularly vulnerable to overharvesting. One such species was the great auk, a large, flightless seabird found on islands in the North Atlantic. By the 1840s, humans had hunted the great auk to extinction to satisfy demand for its feathers, eggs, and meat.

Also susceptible to overharvesting are large organisms with low reproductive rates, such as elephants, whales, and rhinoceroses. The decline of Earth's largest terrestrial animals, the African elephants, is a classic example of the impact of overhunting. Largely because of the trade in ivory, elephant populations have been declining in most of Africa for the last 50 years. An international ban on the sale of new ivory resulted in increased poaching (illegal hunting), so the ban had little effect in much of central and eastern Africa. From 2006 to 2015, the number of elephants dropped by 110,000, a 22% decline (from 525,000 to 415,000). Only in South Africa, where once-decimated herds have been well protected for nearly a century, have elephant populations been stable or increasing (see Figure 53.8). If such protections are not extended to other regions, elephants could become extinct in the wild across most of Africa.

Conservation biologists are increasingly using the tools of molecular genetics to track the origins of tissues harvested from endangered species. For example, researchers have used DNA isolated from samples of elephant dung to construct a DNA reference map for the African elephant (*Loxodonta africana*). By comparing this reference map with DNA isolated from ivory harvested legally or by poachers, they can determine to within a few hundred kilometers where the elephants were killed **(Figure 56.8)**. Such work in Zambia suggested that poaching rates were 30 times higher than previously estimated, a finding that has stimulated improved anti-poaching efforts by the Zambian government. Similarly, biologists using phylogenetic analyses of mitochondrial DNA (mtDNA) showed that some whale meat sold in Japanese fish markets came from illegally harvested species, including fin and humpback whales, which are endangered (see Figure 26.6).

Figure 56.8

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Ecological forensics and elephant poaching.
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Over 200 severed elephant tusks.

These severed tusks were part of an illegal shipment of ivory intercepted on its way from Africa to Singapore in 2002. DNA-based evidence showed that the thousands of elephants killed for the tusks came from a relatively narrow east-west band centered in Zambia rather than from across Africa.

MAKE CONNECTIONS *Figure 26.6**describes a similar example in which conservation biologists used DNA analyses to compare harvested samples of whale meat with a reference DNA database. How are these examples similar, and how are they different? What limitations might there be to using such forensic methods in other suspected cases of poaching?*

*For suggested answer, see*Appendix A.**

Many commercially important fish populations, once thought to be inexhaustible, have been decimated by overfishing. Demands from an increasing human population for protein-rich food, coupled with new harvesting technologies, such as long-line fishing and modern trawlers, have reduced these fish populations to levels that cannot sustain further exploitation. Until the past few decades, the Atlantic bluefin tuna was considered a sport fish of little commercial value—just a few cents per pound for use in cat food. In the 1980s, however, wholesalers began flying fresh, iced bluefin to Japan for sushi and sashimi. In that market, the fish can now bring over $500 per pound **(Figure 56.9)**. With increased harvesting spurred by such high prices, it took just ten years to reduce the western Atlantic bluefin population to less than 20% of its 1980 size.

Figure 56.9

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Overharvesting.
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Hundreds of dead North Atlantic Bluefin tuna in a Japanese fish market.

Atlantic bluefin tuna are auctioned in a Japanese fish market.

Global Change

The fourth threat to biodiversity, global change, alters the fabric of Earth's ecosystems at regional to global scales. Global change includes alterations in climate, atmospheric chemistry, and broad ecological systems

that reduce the capacity of Earth to sustain life.

One of the first types of global change to cause concern was *acid precipitation*, which is rain, snow, or fog with a pH less than 5.2. The burning of wood and fossil fuels releases oxides of sulfur and nitrogen that react with water in air, forming sulfuric and nitric acids. The acids eventually fall to Earth's surface, where they cause chemical reactions that decrease nutrient supplies and increase concentrations of toxic metals. These changes to the soil and water harm some terrestrial and aquatic organisms.

In the 1960s, ecologists determined that lake-dwelling organisms in eastern Canada were dying because of acid precipitation caused by air pollution from factories in the Midwestern United States. Newly hatched lake trout, for instance, die when the pH drops below 5.4. Similarly, lakes and streams in southern Norway and Sweden were losing fish because of pollution generated in Great Britain and central Europe. By 1980, the pH of precipitation in large areas of North America and Europe averaged 4.0–4.5 and sometimes dropped as low as 3.0. (To review pH, see Concept 3.3.)

Environmental regulations and new technologies have enabled many countries to reduce sulfur dioxide emissions in recent decades. In the United States, sulfur dioxide emissions decreased more than 75% between 1990 and 2013, gradually reducing the acidity of precipitation **(Figure 56.10)**. However, ecologists estimate that it will take decades for aquatic ecosystems to recover. Meanwhile, emissions of nitrogen oxides are increasing in the United States, and emissions of sulfur dioxide and acid precipitation continue to damage forests in central and eastern Europe.

Figure 56.10

Changes in the pH of precipitation at the Hubbard Brook Experimental Forest, New Hampshire.

A graph plots p H from 4.0 to 4.9 versus year from 1960 to 2015. p H starts at 4.3 in 1964, drops to 4.0 in 1970 and gradually rises to 4.9 in 2015.

Year from 1960 to 2015 is on the x-axis with increments every 5 years and p H is on the y-axis from 4.0 to 4.9 with increments every 0.1. p H starts at 4.3 in 1964, drops to 4.0 in 1970, rises to 4.4 in 1984, remains relatively steady until 2000 and then gradually rises to over 4.8 in 2010. Between 2010 and 2015 it peaks at over 4.9, then falls slightly by 2015 to 4.9.

MAKE CONNECTIONS *Describe the relationship between* pH *and acidity. (See* Concept 3.3.) Overall, is the precipitation in this forest becoming more acidic or less acidic?*

For suggested answer, see Appendix A.*

We'll explore the importance of global change for Earth's biodiversity in more detail in Concept 56.4, where we examine such factors as climate change and ozone depletion.

Can Extinct Species Be Resurrected?

To the best of our knowledge, extinction has always been forever. Some scientists are nevertheless trying to use cloning techniques (see Concept 20.3) to resurrect species that have become extinct. For example, researchers used this approach in an attempt to reverse the extinction of the Pyrenean ibex, *Capra pyrenaica pyrenaic*a. In 1999, researchers removed a skin sample from the last living individual, a female, and froze it. When that individual died a year later, scientists used the frozen cells to try to produce more ibex. Eventually, one ibex was born in 2009. Sadly, it lived for only 7 minutes before succumbing to lung defects similar to those observed in other cloned animals.

More recently, researchers completed an early step toward the possible use of cloning to prevent the extinction of the northern white rhinoceros (*Ceratotherium simum cottoni*). The last *C. simum cottoni* male died in 2018—dooming the northern white rhinoceros to eventual extinction. That same year, researchers successfully created hybrid embryos using sperm from northern white rhinoceroses and eggs from closely related southern white rhinoceroses, *C. simum simum* (the few living females from the northern subspecies all had severe reproductive issues).

Many steps remain before the northern white rhinoceros could be considered resurrected from extinction. Moreover, even if scientists eventually can resurrect one or a few species, that approach will not solve the extinction crisis we face today. To do that, we must take steps to clean up the environment and protect natural habitat for the thousands of species currently threatened by extinction.